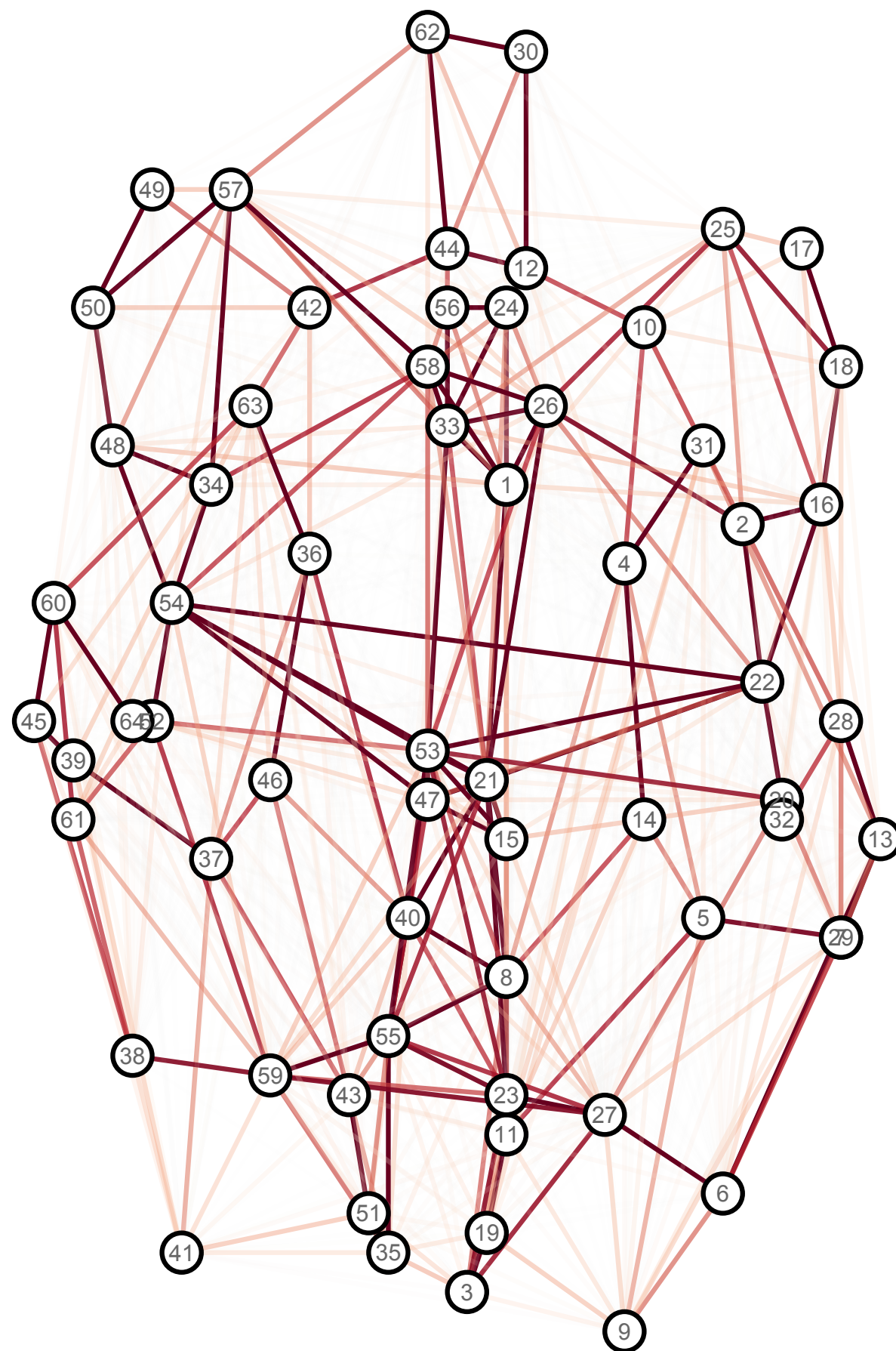




MODELLING THE RELATIONSHIP BETWEEN STRUCTURAL AND FUNCTIONAL CONNECTOMES

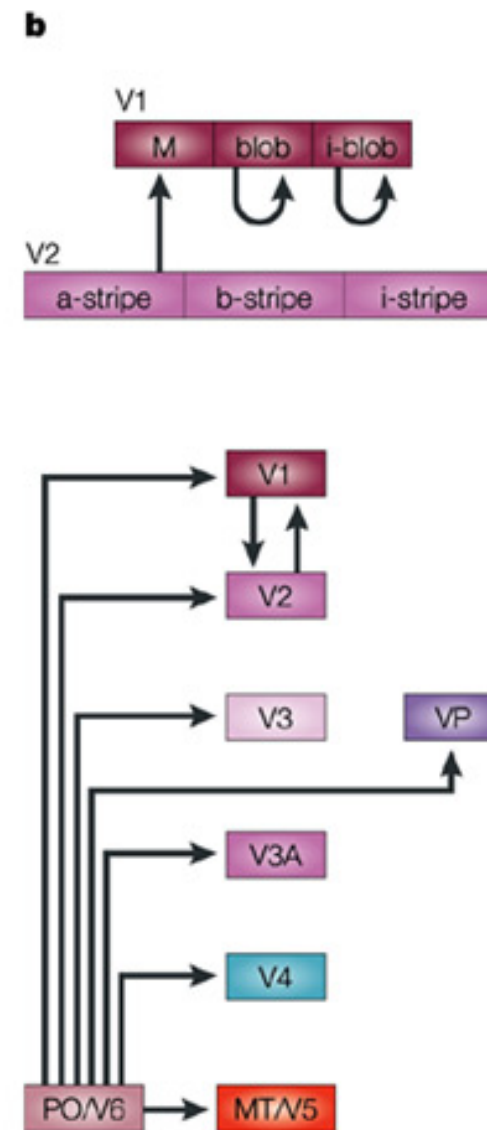
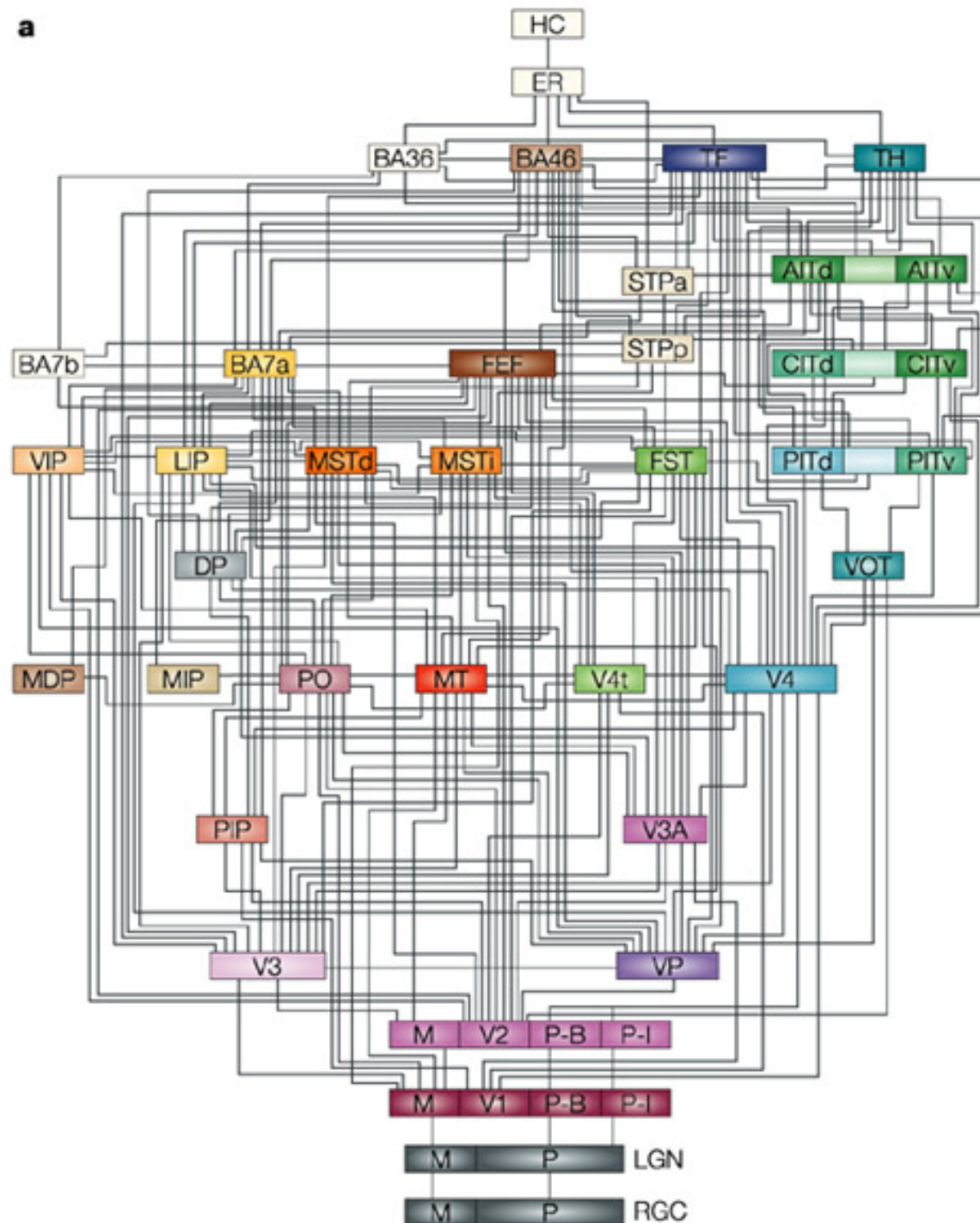
Jon Clayden
j.clayden@ucl.ac.uk



AIMS

- Understand the concepts of structural and functional connectivity
- Understand how they can be estimated using biomedical imaging
- Investigate the effects of image processing decisions on connectivity networks
- Explore how the two forms of connectivity are related, and appreciate why there may be a mismatch
- Explicitly model these relationships
- Vary models and assumptions, and compare outcomes

CONNECTIVITY AND THE BRAIN

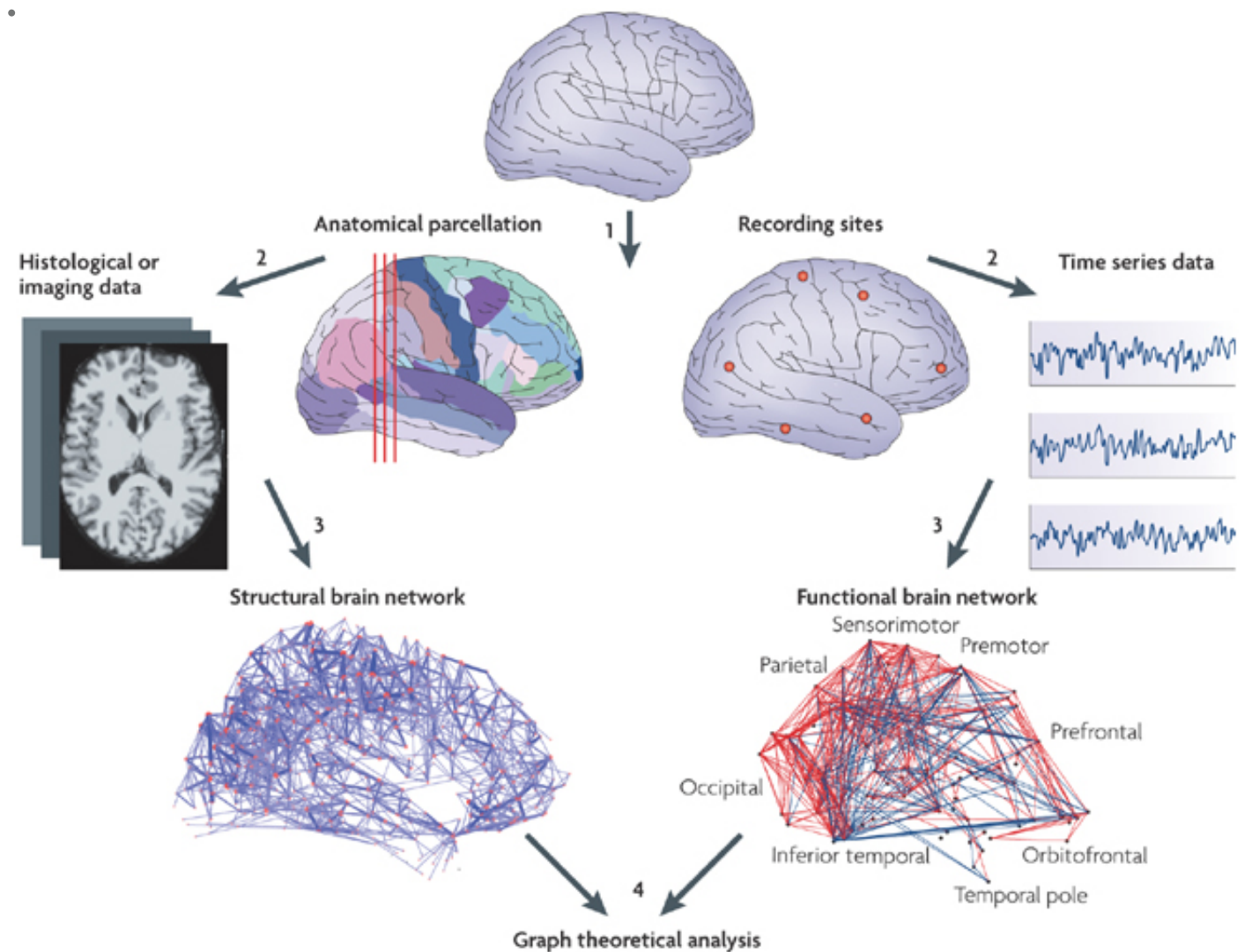


Rees et al., *Nat Rev Neurosci*,
2002 (after Felleman & van
Essen, *Cereb Cortex*, 1991)

DEFINITIONS OF CONNECTIVITY

.....

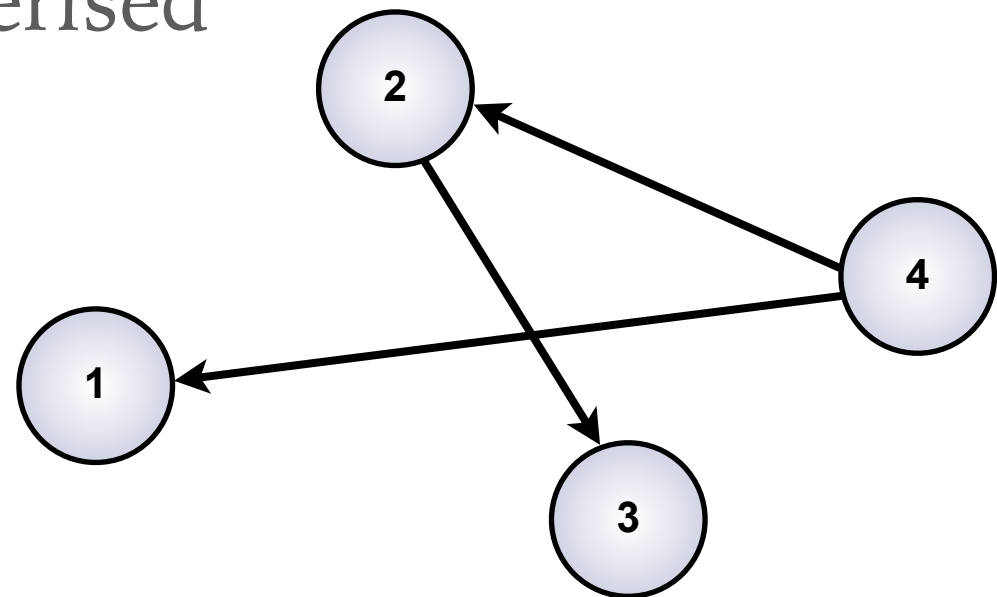
- **Structural connectivity:**
the physical axon bundles connecting brain regions together
- **Functional connectivity:**
associations between neural activity in spatially remote regions of grey matter
- **Effective connectivity:**
patterns of influence by some neural systems over others



Bullmore & Sporns, *Nat Rev Neurosci*, 2009

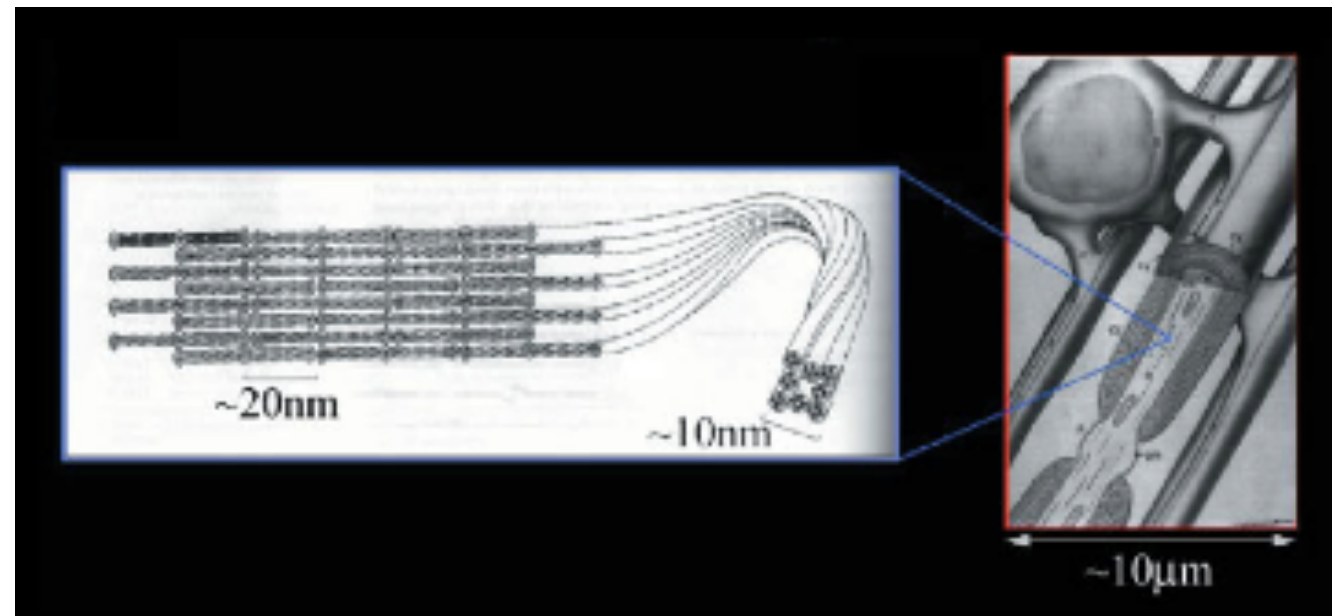
GRAPHS

- A highly abstract representation of a set of **vertices** connected by **edges**
- Edges may be **directed** or **undirected**, and may have associated **weights** or costs
- A natural representation of connected systems
- Theoretically very well characterised
- Broad range of applications



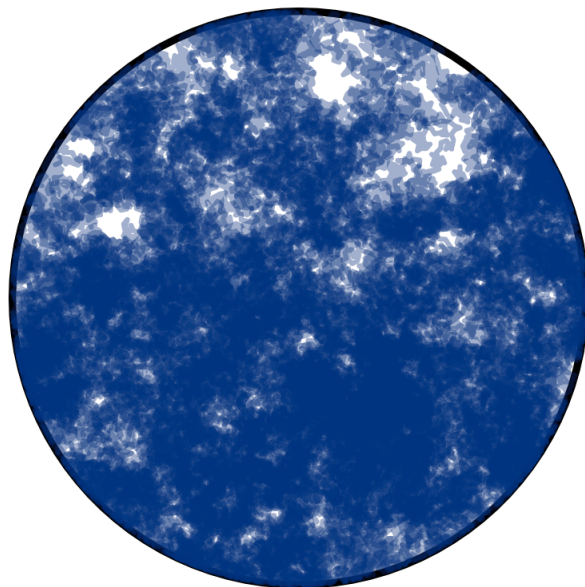
DIFFUSION ANISOTROPY IN TISSUE

.....

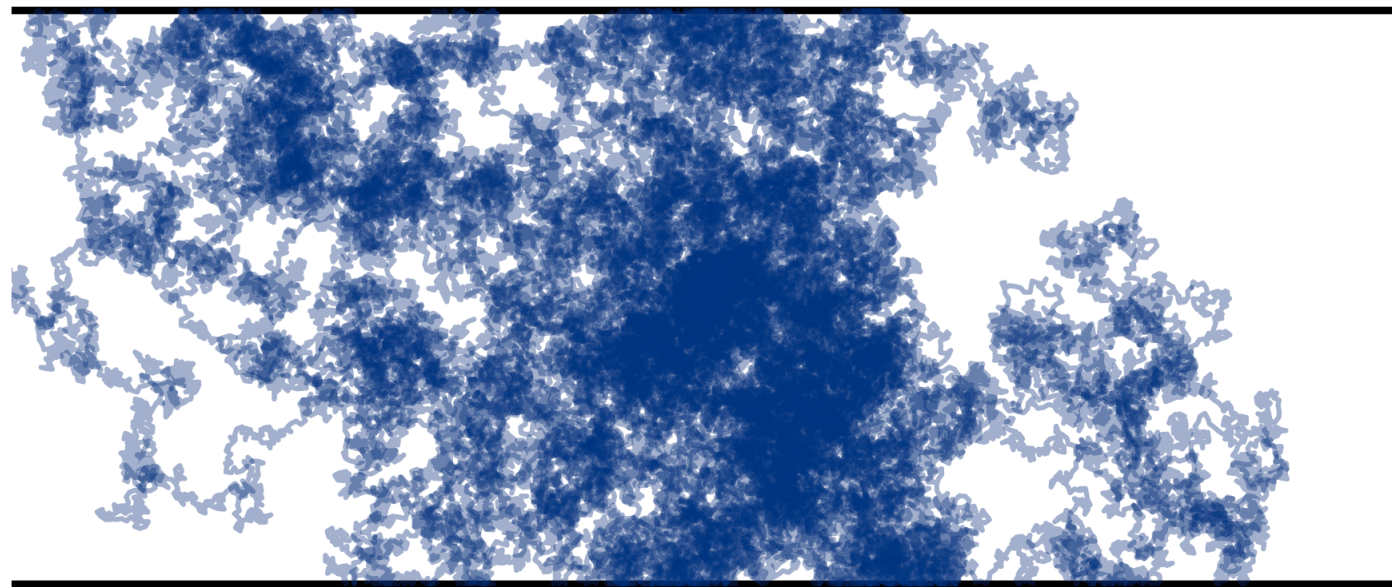


*Mori & van Zijl,
NMR Biomed,
2002*

A

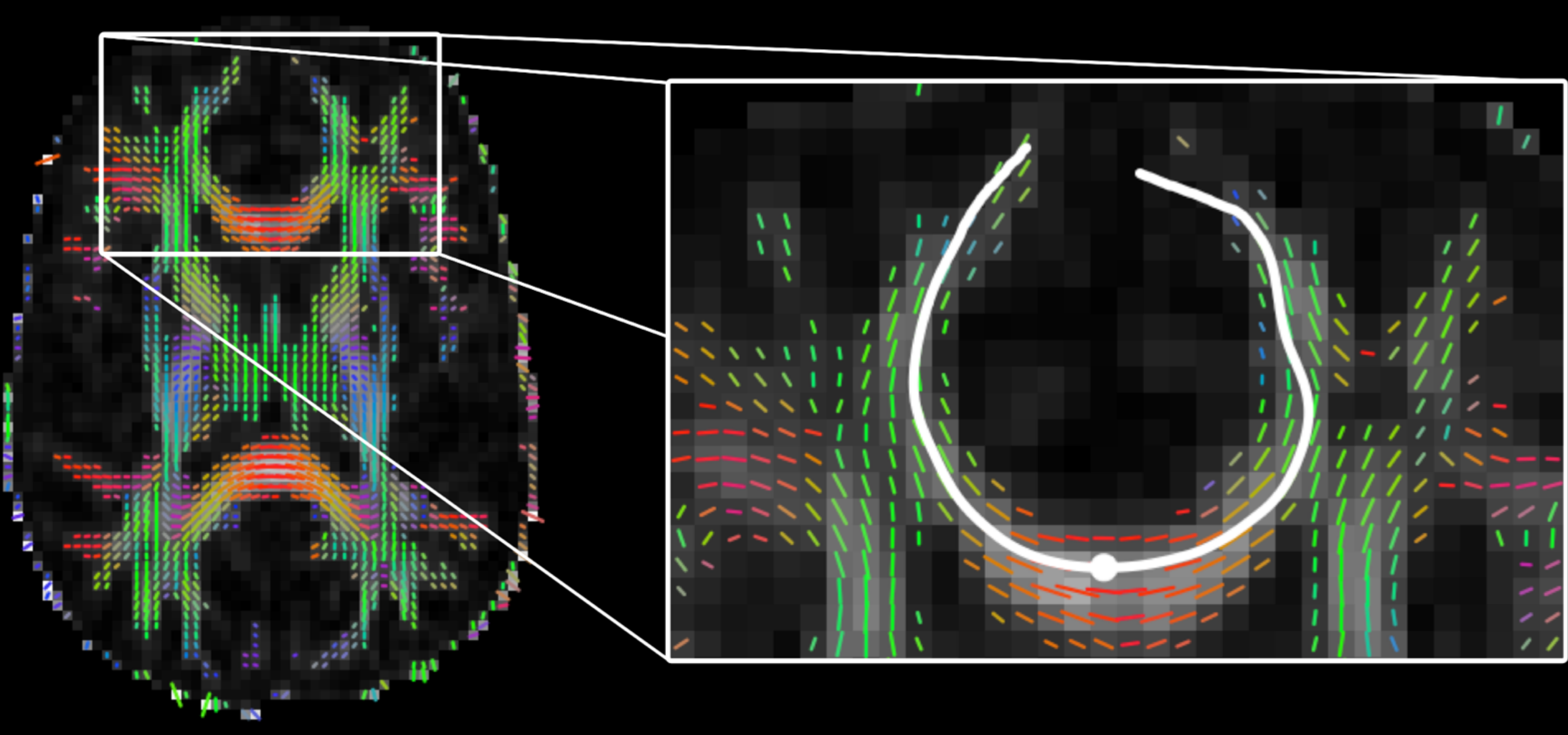


B



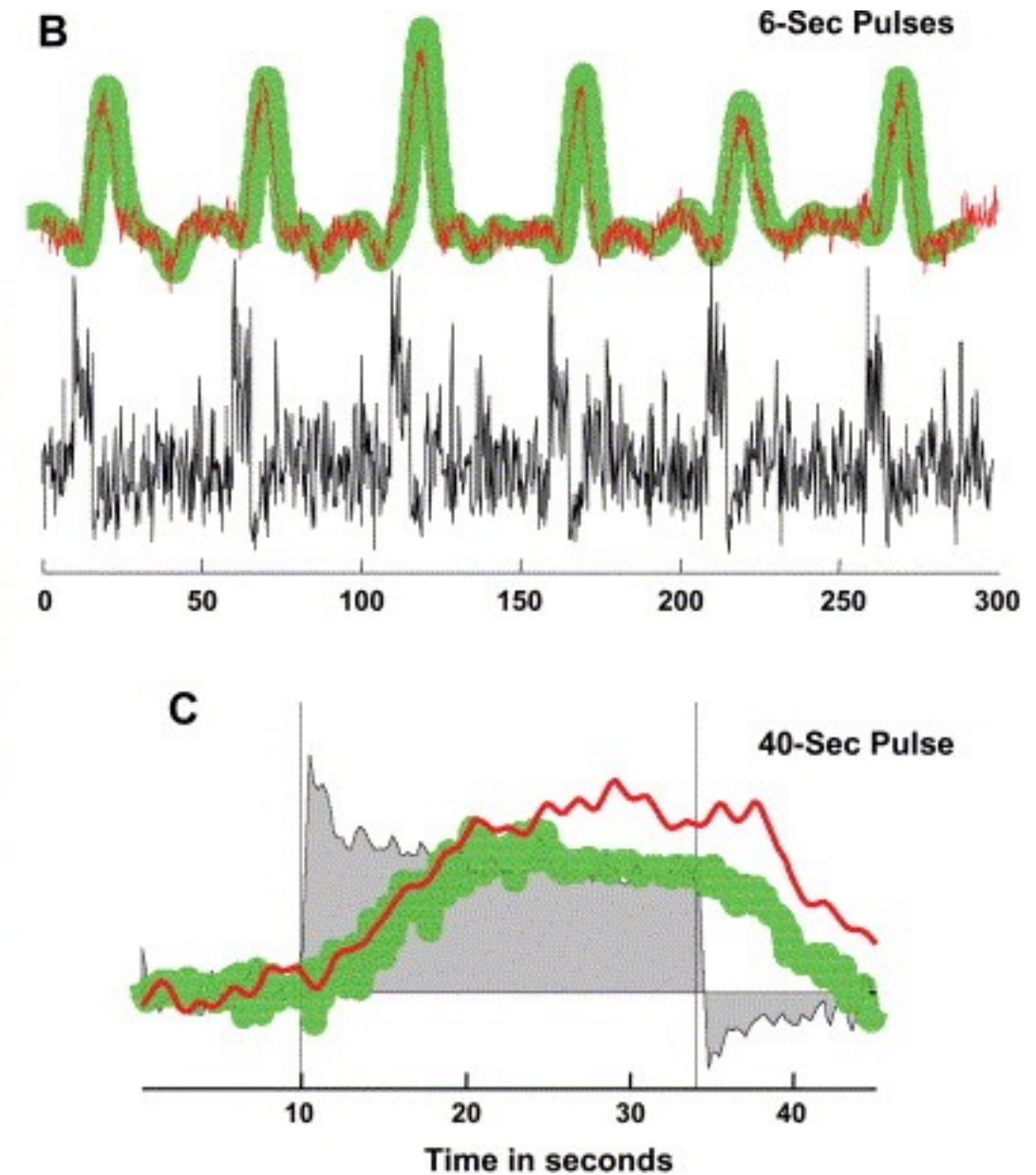
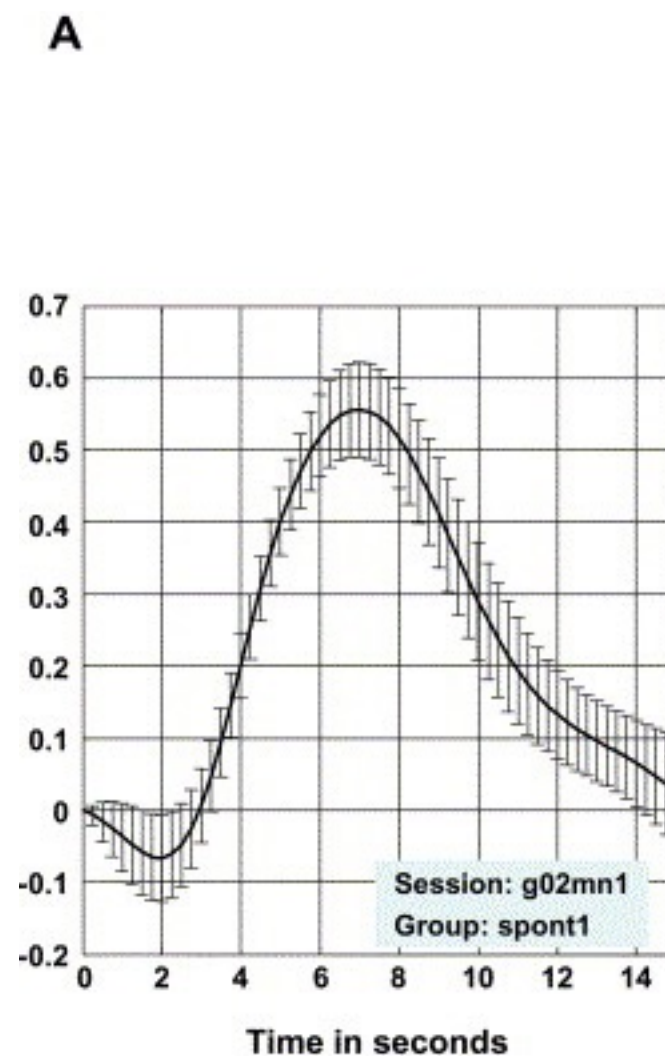
STRUCTURAL CONNECTIVITY: TRACTOGRAPHY

.....



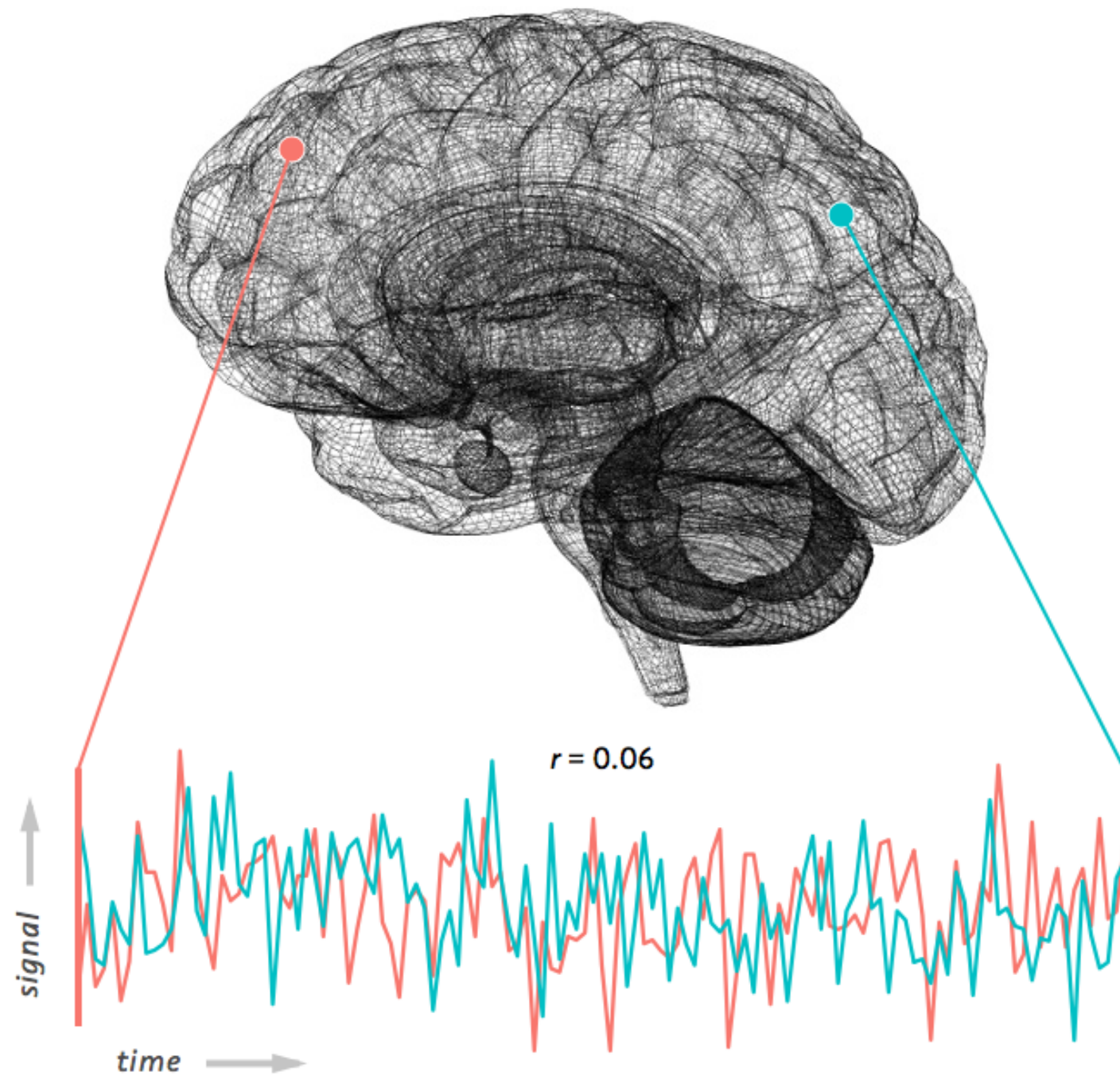
FUNCTIONAL MRI

- Brain activity linked to increased demand for oxygenated blood
- Associated via a **haemodynamic response function**
- Functional activity estimated by **deconvolution**

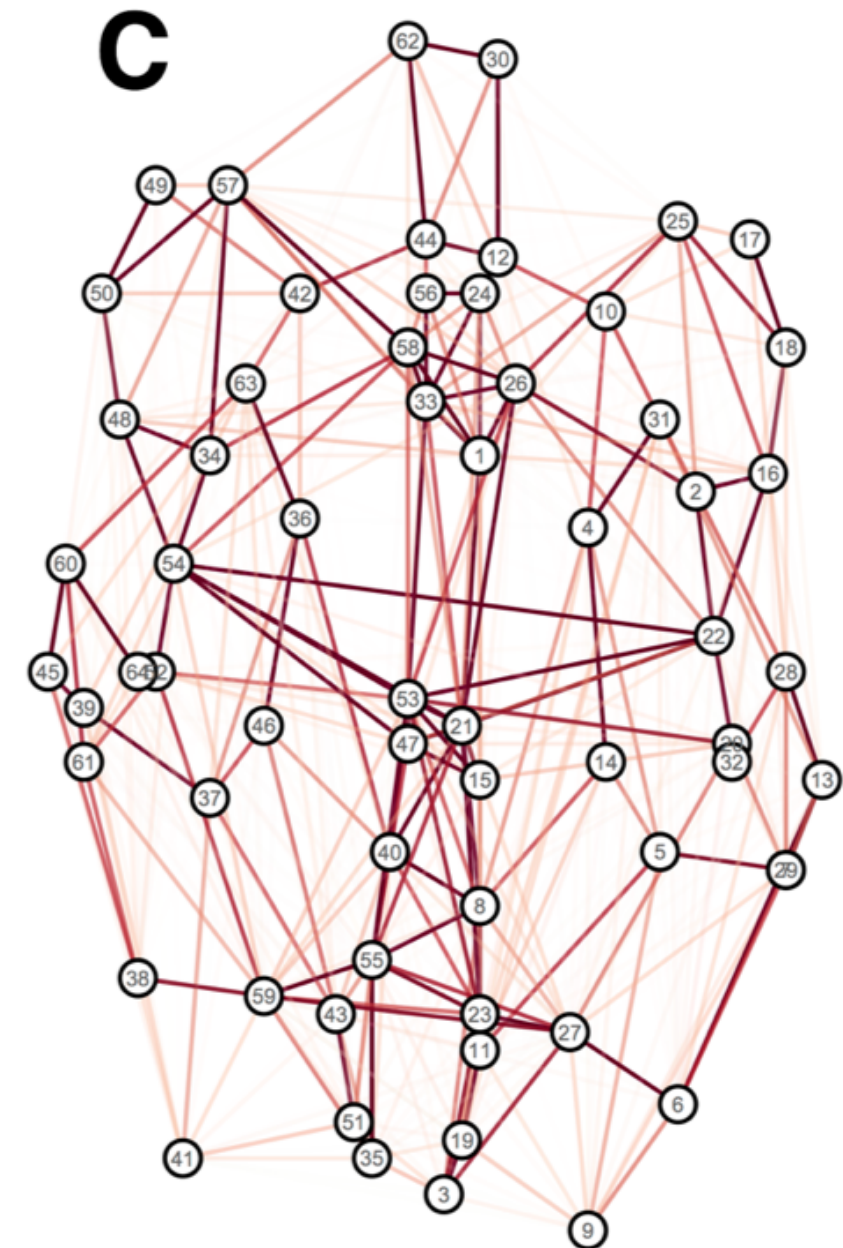
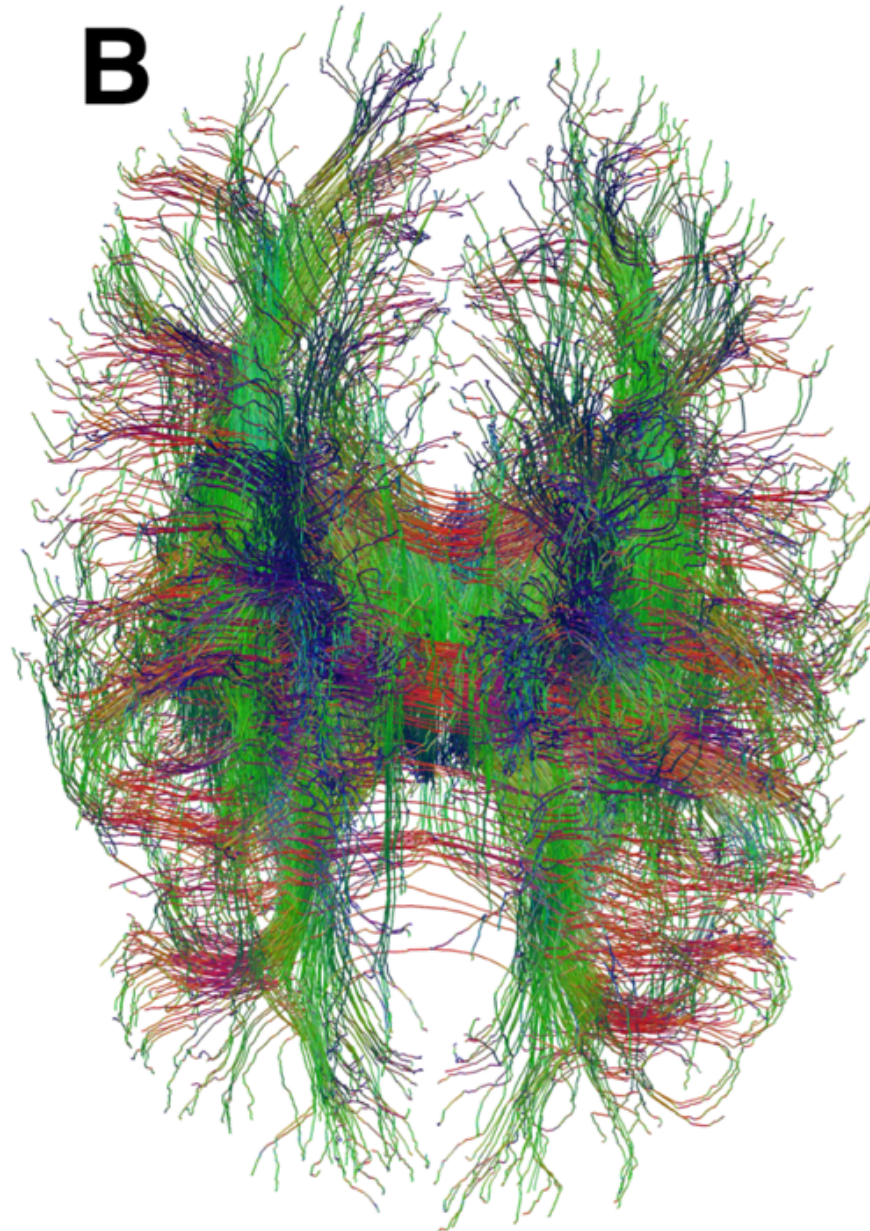
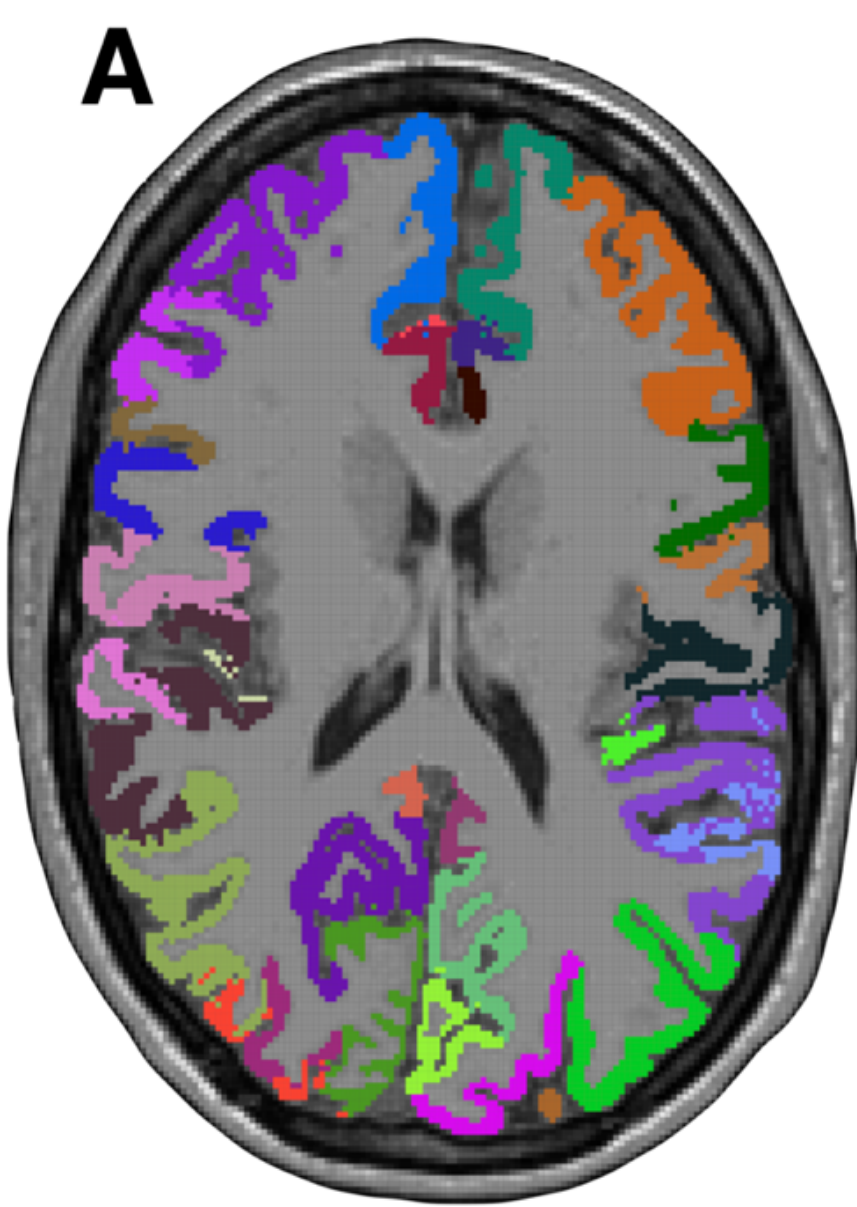


Logothetis & Pfeuffer, Magn Reson Imag, 2004

FUNCTIONAL CONNECTIVITY: CORRELATED TIME-COURSES



IN VIVO CONNECTOMICS

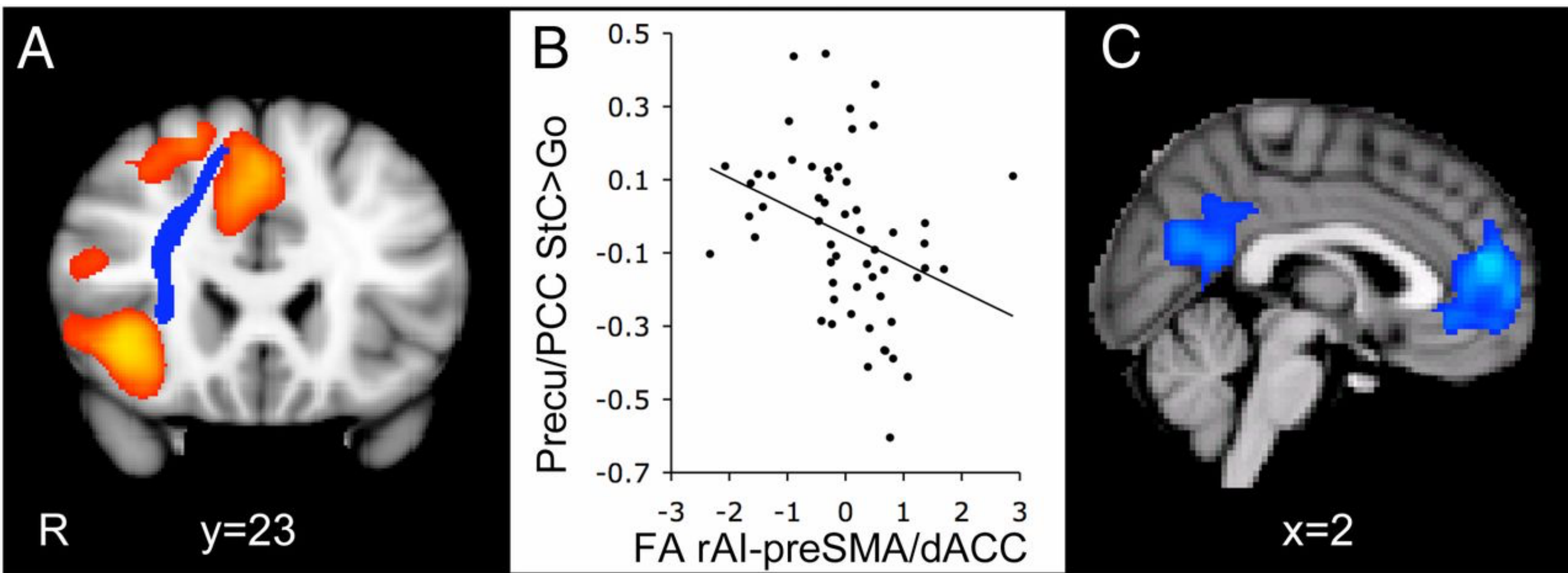


GRAPH CHARACTERISTICS

- A range of measures have been developed in graph theory to describe characteristics of graphs and their vertices
- **Connection density**: the proportion of all possible edges which are present in the graph (cost)
- Average **path length**: the mean shortest path length between pairs of vertices (efficiency)
- **Betweenness centrality**: the number of shortest paths between other vertices which pass through a particular vertex (hubs)

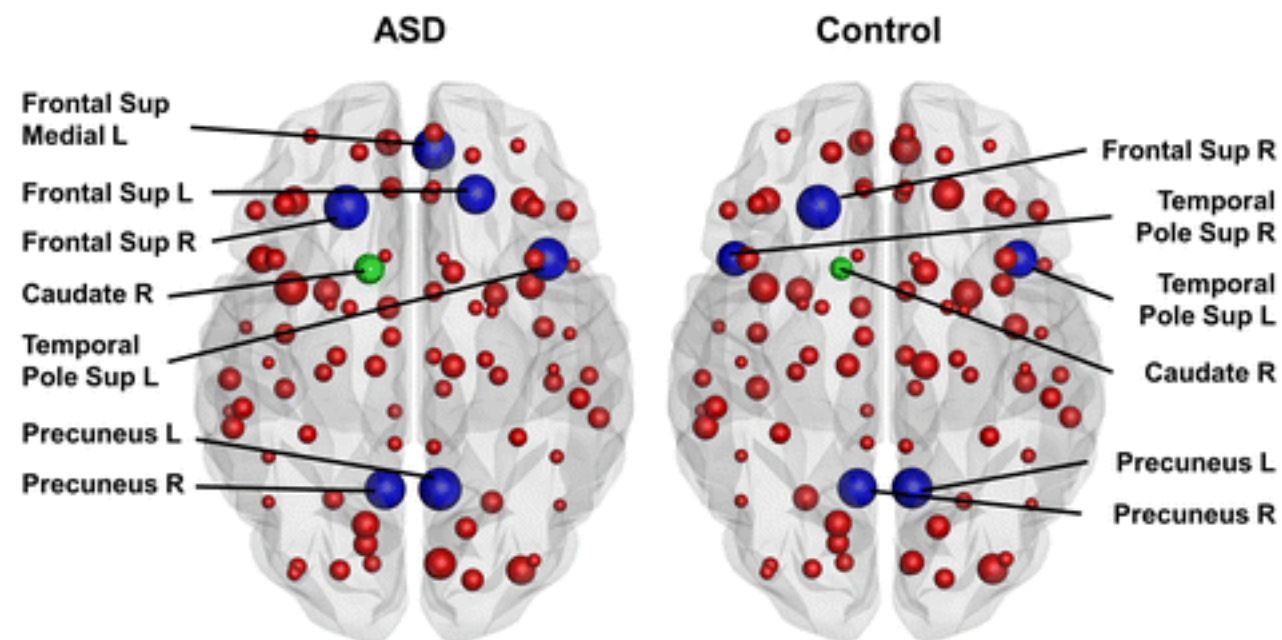
STRUCTURE–FUNCTION RELATIONSHIPS

- Integrity of a particular white matter tract predicts de-activation of the **default-mode** functional network after injury



APPLICATION TO ASD

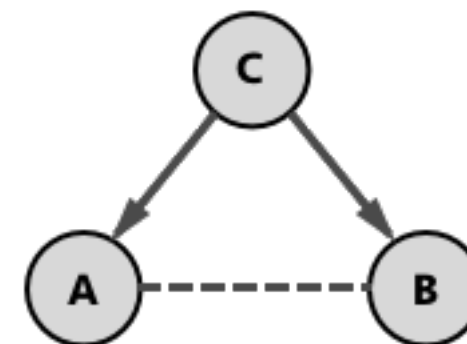
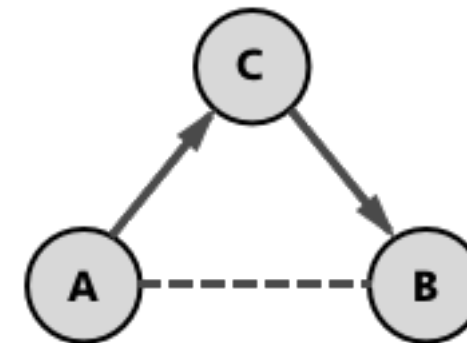
- Global network differences between **ASD** and control groups, e.g. in network efficiency
- Also local differences in centrality and hub structure



Roine et al., *Mol Autism*, 2015

INDIRECT CONNECTIONS

- Correlated activity may indicate a **direct** physical connection ...
- ... or an **indirect** one ...
- ... or top-down influence from a third region



OUTLINE OF THE TASKS

- Construct a functional connectome from an individual dataset
- Explore the effect of certain parameters on the graph-theoretical properties of structural and functional connectomes
- Directly model the relationship between structural connection weights and the equivalent functional connection weights
- Incorporate indirect structural connections, and investigate their influence
- Predict functional connectomes from structural ones, and compare different models for doing so